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Selective Breeding of *Penaeus vannamei*: Impact on World Aquaculture and Lessons for Future

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ABSTRACT

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Shrimp farmers want fast growing shrimp that can resist common diseases and get good survivals in pond culture. The SPF *Penaeus vannamei* have produced these results and most of Asia has switched to farming SPF *P. vannamei*. Downward price pressure in the market is causing farmers to demand still better performance to remain profitable. The best opportunity to improve economic performance in shrimp farming is through selective breeding of the commercial shrimp stocks. Three different approaches to shrimp breeding are being employed globally. First is the SPF model, undertaken by US-based breeding companies who supply most of Asia's shrimp broodstock. SPF stocks selected for fast growth have resulted in significant improvements up to 10% gain per generation. Average daily gain (ADG) now exceeds 0.3 gram per day (gpd). In farms stocked at more than 100 PL/m2, farmers harvest 20 g shrimp in less than 100 DOC. These results have propelled the industry forward and India is one of the largest beneficiaries of this approach. A second approach to breeding shrimp is the Ecuador model. Ecuador is the leading shrimp farming country in the Western Hemisphere. Following an industry crash caused by White Spot Virus in 1999, the Ecuador industry initiated a breeding program based on collecting survivors of pond mortality events and over time created a stock of shrimp that are genetically tolerant to WSSV infection. While these efforts ignored the growth characteristics of the shrimp, pond survival rates steadily increased generation by generation and the industry not only recovered from the WSSV problems, it flourished with production of 500,000 MT in 2018. A third model for shrimp breeding combines both approaches. It has been called the SPF/SPT model (Alday-Sanz *et al*., 2018). This approach starts with a pathogen-tolerant stock as founding stock, passes them through an SPF development process and results in an SPF/SPT stock that are both SPF (pathogen free) with pathogen-tolerant genetics.

ADDITIONAL INDEX WORDS: Penaeus vannamei, *specific pathogen free stock, shrimp culture, selective breeding.*

INTRODUCTION

Pacific white shrimp (*Penaeus vannamei*) is native to the tropical Pacific coast of Latin America and has been the species of choice for shrimp farming in the Americas since the early 1970s. Up to 1990, *P. vannamei* contributed less than 10% of total world production of farmed shrimp while black tiger shrimp (*P. monodon*), used by Asian shrimp farmers, contributed more than 90% of total world production. Development and industry-wide use of specific pathogen free (SPF) *P. vannamei* in the US in the early 1990s resulted in a doubling of US industry production (Figure 1). Success with SPF *P. vannamei* led to development of selective breeding programs and commercial shrimp breeding companies in the US to capitalize on these promising results. Breeding programs applied family-based selection for disease resistance combined with selection for fast growth. Several groups achieved rapid progress developing Taura Virus resistance and improved growth rates of selected shrimp. These improvements in shrimp performance combined with *P. vannamei*'s desirable natural traits advanced shrimp farming production and set the stage for globalization of this species.

The SPF *P. vannamei* was first introduced to Asia in the late 1990s and produced dramatic increases in production over P. monodon that led to rapid spread throughout Southeast Asia. Sustained increases in Asian shrimp production resulted from *P. vannamei*'s widespread adoption and resulted in a tripling of global shrimp production between 2000 and 2010. By 2010, *P. vannamei* production accounted for 80% of total world production and was the dominant species farmed in India, China, Indonesia and Vietnam-the world's leading production countries (Figure 2).

Since the introduction of SPF *P. vannamei* to Asia, global shrimp farming annual crop value also tripled. In 1997, global farmed-shrimp annual production of 700,000 MT had a total crop value of about $3.5 billion (FAO, 2010). Widespread adoption of *P. vannamei* over the next 10 years resulted in sustained production increases of more than 10%/yr. While shrimp prices declined as a result of this increasing supply, total shrimp crop value in 2010 was worth $18.5 billion based on production of 3.7 million MT (FAO, 2012). This 370% increase in industry crop value in 13 years is the result from domestication and globalization of *P. vannamei*. The SPF *P.*

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vannamei introduction and adoption in Asia added $224 billion in shrimp crop value since 2000.

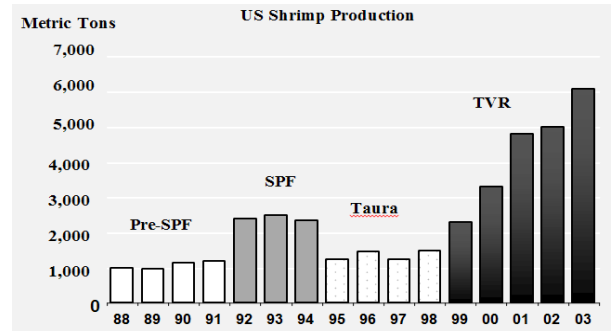


Figure 1. US shrimp farming industry annual production in metric tons divided into four eras: Pre-SPF (1988-91), SPF ('92-94), Taura ('95-98) and TVR ('99-2003) (after Wyban, 2016).

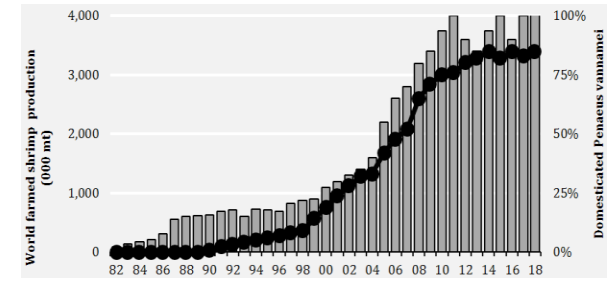


Figure 2. World shrimp farming annual production (000 MT) and percent contribution to total by domesticated *Penaeus vannamei* (Wyban, 2016).

Breeding SPF *P. vannamei* for Fast Growth and Virus Resistance

To develop fast-growing shrimp with good resistance to common diseases, High Health Aquaculture (HHA) used performance testing of multiple families of its shrimp lines each generation. We used a combined selection strategy based on shrimp survival in replicated laboratory virus challenges combined with shrimp growth performance under simulated commercial conditions. From 1996, HHA employed this combined selection approach to develop a fast-growing, TSV-resistant stock, we called GxTVR.

TSV and IMNV are viruses that cause significant economic disease in farmed shrimp. Herein, we describe a recent performance trial comparing growth rates and virus challenge results for a diverse collection of SPF shrimp. Four lines of shrimp, three *P. vannamei* and one *P. stylirostris*, were compared. Multiple same-age families represented each line. All families were tested in two virus challenges (TSV and IMNV) at the Shrimp Pathology Group in Tucson, Arizona (White *et al*., 2002). Equivalent tagged samples of the same families were tested in a growth trial at HHA in Kona Hawaii.

A collection of 12 same-age (*i.e.*, spawned the same day) SPF shrimp families was produced in High Health Aquaculture's Shrimp Breeding Center in Kona Hawaii. The twelve families were derived from four distinct genetic lines as listed in Table 1.

Table 1. *Composition of four shrimp groups in virus challenges and growth trial.*

Species	Line	Families
P. vannamei	GxTVR	4
P. vannamei	Hybrid	4
P. vannamei	Kona Stock	2
P. stylirostris	SPF Blue Shrimp	2

GxTVR is HHA's commercial stock that has been selected for fast growth and TSV resistance for 14 generations. Families from this stock routinely achieve >90% survival in TSV lab challenges. Kona Stock is the original SPF shrimp stock developed in 1990. It is routinely used as the positive control stock in TSV challenges because of its well-documented susceptibility to TSV. The HHA has maintained this SPF stock for this purpose since 1994. The Hybrid Line was produced by mating GxTVR and Kona Stock parents. HHA also breeds SPF Blue Shrimp (*P. stylirostris)* since 1994. Blue shrimp are naturally resistant to TSV and we wanted to determine their relative resistance to IMNV.

The twelve families were produced at HHA's shrimp breeding center in Kona Hawaii and color-code tagged with elastomer by injection. Samples of the tagged families were shipped to the Shrimp Pathology Group (SPG) in Tucson, Arizona for virus challenge testing (White *et al.,* 2005). SPG was contracted by HHA to conduct the virus challenges using purified strains of both viruses. SPG recently developed a lab challenge bioassay with purified IMNV and we tested it with HHA shrimp. At SPG, the lines were challenge-tested with two different virus isolates, IMNV and TSV. Both positive and negative control treatments were included in the experimental design. IMNV challenge was administered by injection of a viral preparation in the 3rd abdominal segment of each test shrimp. TSV challenge was achieved by feeding (*per os*) TSV-contaminated shrimp tissue to the challenged animals. Mortalities in both challenges were recorded daily and representative samples of moribund shrimp were examined to determine cause of death. A growth trial was conducted at HHA in Hawaii using a parallel set of tagged individuals from the same families and lines as the virus challenges. The growth trial shrimp were tagged using the same color scheme as the viral challenges. The shrimp were stocked into a single tank at HHA at 125 shrimp per m^2. Shrimp were reared under a low water exchange, high aeration system using commercial diet. Shrimp were sampled monthly with 50 pcs per family individually scored to family and weighed.

In the virus challenges, survival by shrimp line is shown in Figure 3. In the both IMNV and TSV challenges, GxTVR line had the highest survival among the *P. vannamei* lines. Kona Stock shrimp had the lowest survival while the hybrids were intermediate. Blue shrimp survival was not different from GxTVR stock in both IMNV and TSV challenges. It was surprising to see a very similar pattern of susceptibility to IMNV as in TSV across all four strains. Are TSV and IMNV susceptibility and resistance a genetically related trait? A scatter gram of family survival in TSV *v.s.* IMNV shows that these two traits were highly correlated with $r^2 = 0.84$ (Figure 4). This high

correlation among the families suggests there may be some common genetic factor for both viruses which requires further study. In other challenge studies, we saw no correlation between TSV and WSSV susceptibility.

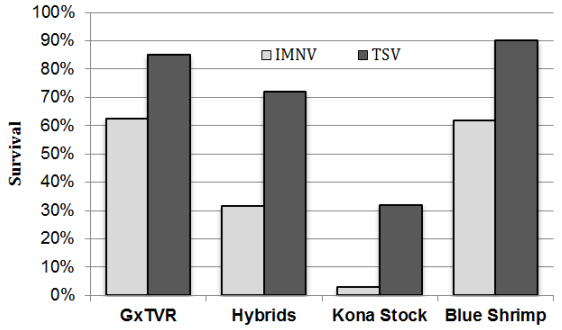


Figure 3. Virus challenge survivals of 4 shrimp lines with TSV and IMNV (Wyban, 2012).

Taura Syndrome (TS) is a widespread viral disease in both hemispheres. Results in the challenge study with the Thai strain of TSV were similar to our previous studies. GxTVR families were more resistant to TSV than Kona Stock or Kona Stock X GxTVR hybrids. These results clearly demonstrate that selective breeding for TS-resistance has been highly successful. Blue Shrimp were also TSV resistant. There were no benefits in hybrids for TSV resistance. An understanding of why the hybrid families were intermediate to the resistant and susceptible stocks is intriguing. Perhaps the genetics of TSV is a simple Mendalian gene or it is polygenic. Further, molecular studies to understand this variation are needed. Farmers operating where TSV is established should use shrimp stocks that carry documented TSV resistance.

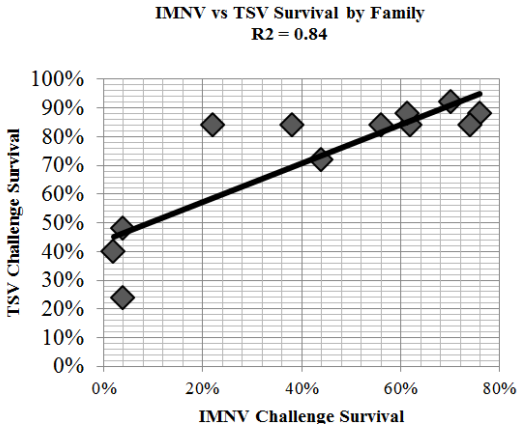


Figure 4. Regression analysis comparing four stocks survival in challenges with TSV and IMNV (Wyban, 2012).

In the growth trial, the four strains were very similar up to 15 g and then separated in growth to adult size with Blue Shrimp growing to largest size followed closely by GxTVR then hybrids and finally Kona Stock (Figure 5). These results are consistent with previous experience at HHA and the breeding history of these stocks. We consistently see Blue Shrimp grow to larger sizes than *P. vannamei*. There are reports that shrimp progeny from matings between genetically distinct groups (hybrid crossing) exhibit faster growth (*i.e.* heterosis) than either of the parent stocks. Since our GxTVR line and Kona Stock line are distinct genetic lines, we wanted to determine if hybrids between them would outperform the parent stocks. In both virus challenges and the growth trial, the hybrids were intermediate to the two pure stocks. For better growth performance, farmers should seek shrimp suppliers with documented growth rate breeding programs.

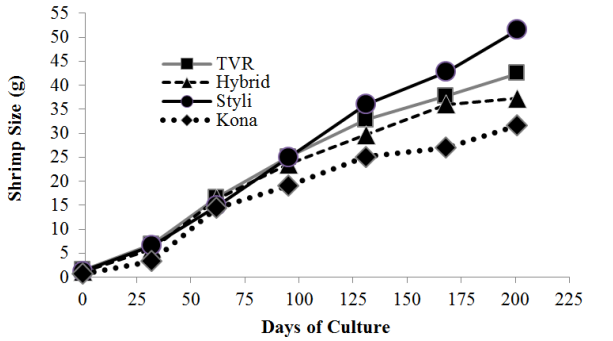


Figure 5. Shrimp growth rates of four lines stocked at 150 pcs/m^2 (Wyban, 2012).

Shrimp farmers want fast-growing, disease-resistant seed. To produce these, shrimp breeding companies can use performance testing to determine best stocks for breeding for desirable traits. This experiment illustrates our approach to such a challenge. Multiple shrimp families from distinct strains were compared for their resistances to two important viruses and their growth rates.

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World shrimp production using *P. vannamei* is now more than 80% of total world production. In this last decade, *P. vannamei* production expanded from only 10% of total world production to this dominant position. This transformation is driven by *P. vannamei'*s lower production costs which derive from advancing domestication coupled with *P. vannamei'*s natural growth traits. With this shift to *P. vannamei*, Taura Syndrome and IMNV remain important threat because they can cause huge production losses in *P. vannamei* farming. The *P. vannamei* component of the industry needs a stable supply of domesticated broodstock shrimp that are resistant to both TS and IMNV with reliable commercial performance in pond culture. As the industry

matures, an ever increasing share of production will be based on use of certified stocks that are SPF and disease-resistant. Like other animal production industries, sustainable shrimp farming will depend on domesticated, certified pathogen-free, fast growing, pathogen-resistant stocks. Our results clearly demonstrate that selective breeding for virus resistance and fast growth is achievable and will contribute to the industry's long-term drive to reduce production costs and increase production reliability.

Alternative Shrimp Breeding Models

Ecuador is the leading shrimp farming country in the Americas. They follow a distinctly different model for shrimp farming that uses hatchery-produced seed (PL) stocked at low densities (10-25 pcs/m^2) in large (2-20 ha) ponds that are not aerated. Most of the Ecuador's production is produced by large integrated companies. This approach to shrimp farming has evolved over more than 40 years. In 1999, White spot virus mortalities struck Ecuador and total output plummeted (Figure 6).

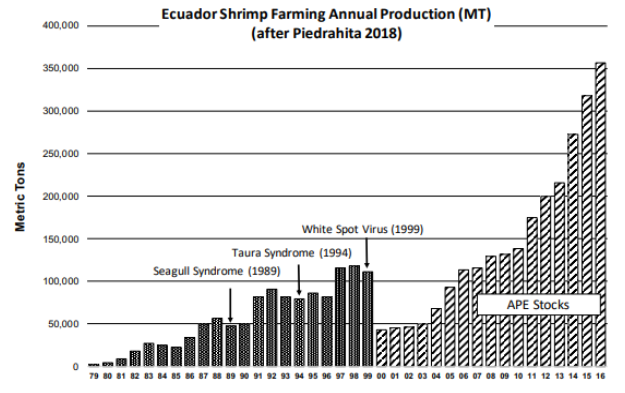


Figure 6. Ecuador annual shrimp production (after Piedrahita, 2018).

In response to this disease crisis, shrimp farming companies began selecting pond survivors following grow-out cycles to become breeding stock for the next generation. By selecting survivors, they were practicing mass selection for resistance or tolerance to WSSV. Year over year, survivals steadily increased and industry output steadily improved. This approach can be called Pathogen Tolerant breeding. The benefits of this approach are clearly illustrated in Ecuador's steady increase in production reaching 500,000 MT by 2018 (Figure 6).

Breeding SPF/SPT Shrimp

Recent developments by 2 different commercial shrimp projects represent a new approach to shrimp breeding that combines the benefits of SPF shrimp (*i.e.*, pathogen free) with the genetics of Tolerant shrimp and holds great promise. In the US, an intensive shrimp farming company in Florida called American Penaeid (API) has developed an SPF stock of *P. vannamei* that was derived from the Ecuador WS-Tolerant shrimp. The company acquired a non-SPF Ecuadorian stock and then passed them through the process required to make them SPF. API is now supplying broodstock to China and results have been impressive. While shrimp growth is slower than other SPF stocks, survivals are much higher, and farmers prefer these animals. A similar strategy was followed by the large shrimp company in Saudi Arabia called National Aquaculture Group (NAQUA). NAQUA acquired non-SPF shrimp stocks from Ecuador that carry significant genetic resistance to WSSV. They then cleaned them up to be SPF and began using this SPF/SPT stock in 2014. Results to date using this shrimp have been impressive (Figure 7).

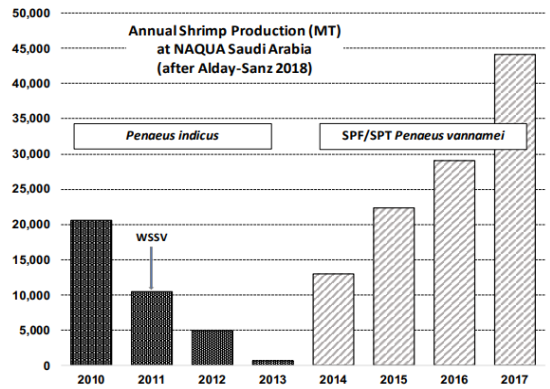


Figure 7. Shrimp production by NAQUA in Saudi Arabia by domesticated *P. vannamei* (from Alday-Sanz *et al*., 2018).

CONCLUSIONS

Each of the three approaches to shrimp breeding described here has benefits and problems. Going forward, shrimp industries will try various combinations of these to try to optimize production economics. It is clear that SPF technology and shrimp breeding are extremely valuable technologies that can make or break a national industry.

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